

README.md

Blogging System API

A secure RESTful blogging backend built with **ASP.NET Core Web API** and **C#**.

Features

- JWT Authentication
- Refresh Tokens
- Email OTP Verification
- Password Reset
- Role-Based Authorization
- Blog Post Management
- Draft Publishing
- Markdown Content
- Threaded Comments
- PostgreSQL Persistence
- Entity Framework Core
- OpenTelemetry Logging
- Swagger/OpenAPI
- Postman Collection

Technology Stack

- C#
 - ASP.NET Core Web API
 - .NET SDK
 - PostgreSQL
 - Entity Framework Core
 - BCrypt
 - JWT
 - SMTP
 - OpenTelemetry
 - Swagger
 - Postman
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Project Structure

```
BlogSystem/  
  
src/  
  Blog.API  
  Blog.Application  
  Blog.Domain  
  Blog.Infrastructure  
  
tests/  
  Blog.UnitTests  
  Blog.IntegrationTests  
  
docs/  
postman/
```

User Roles

Administrator

- Create posts
- Update posts
- Delete posts
- Save drafts
- Publish drafts
- Reply to comments
- Delete any comment
- Lock or unlock comments

User

- Register
 - Verify email
 - Login
 - View posts
 - Comment
 - Reply
 - Delete own comments
 - View profile
 - Update profile
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Running the Project

1. Clone the repository.
 2. Configure PostgreSQL.
 3. Configure SMTP settings.
 4. Update `appsettings.json` (or environment variables).
 5. Apply Entity Framework Core migrations.
 6. Run the API.
 7. Open Swagger.
 8. Test endpoints using Postman.
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Documentation

Project documentation is located in the `docs/` directory.

API Documentation

Swagger/OpenAPI is available when the application is running.

Testing

The project shall be tested using:

- `dotnet build`
 - `dotnet test`
 - Swagger
 - Postman
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License

This project is intended for educational purposes and as a portfolio demonstration of modern ASP.NET Core backend development practices.